

Schematic V1.0

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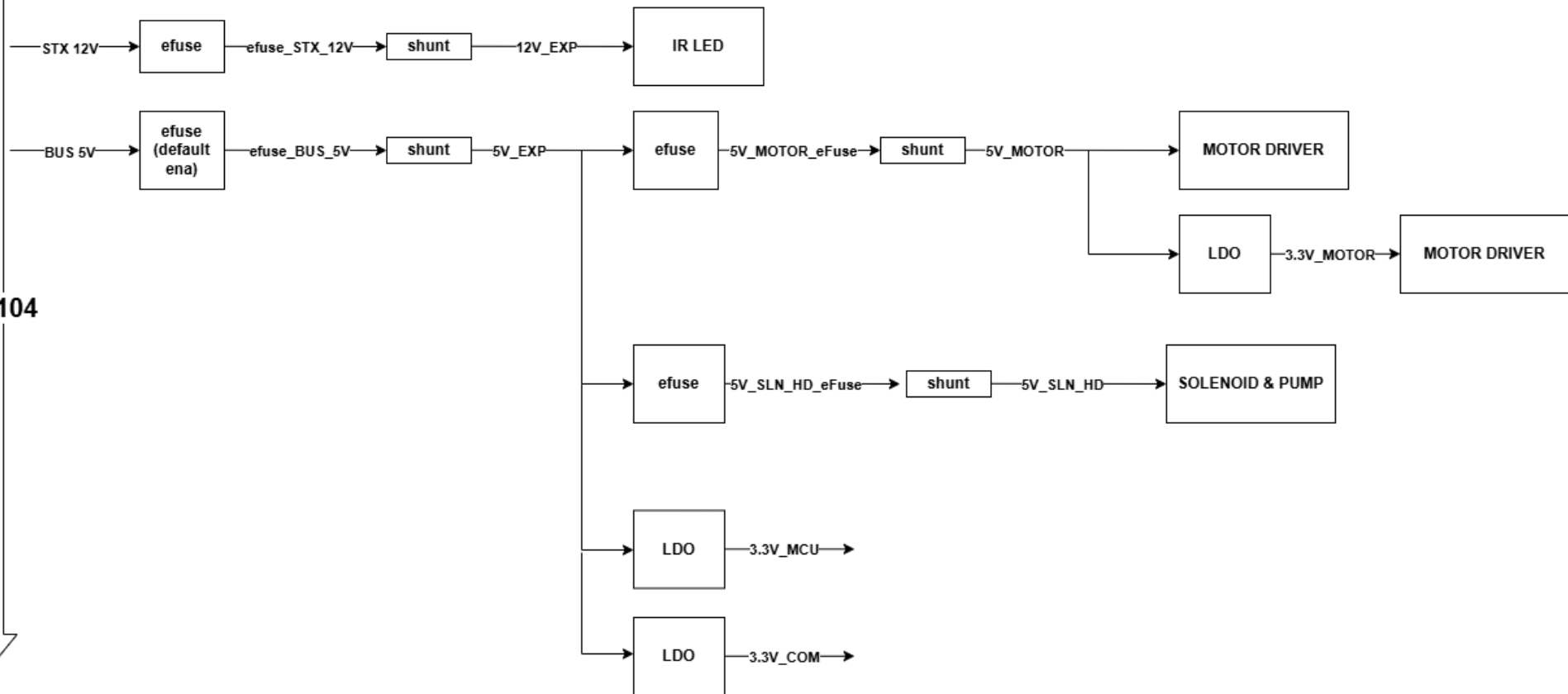
History

VERSION	DISCRIPTION
1.0	INITIAL PROJECT



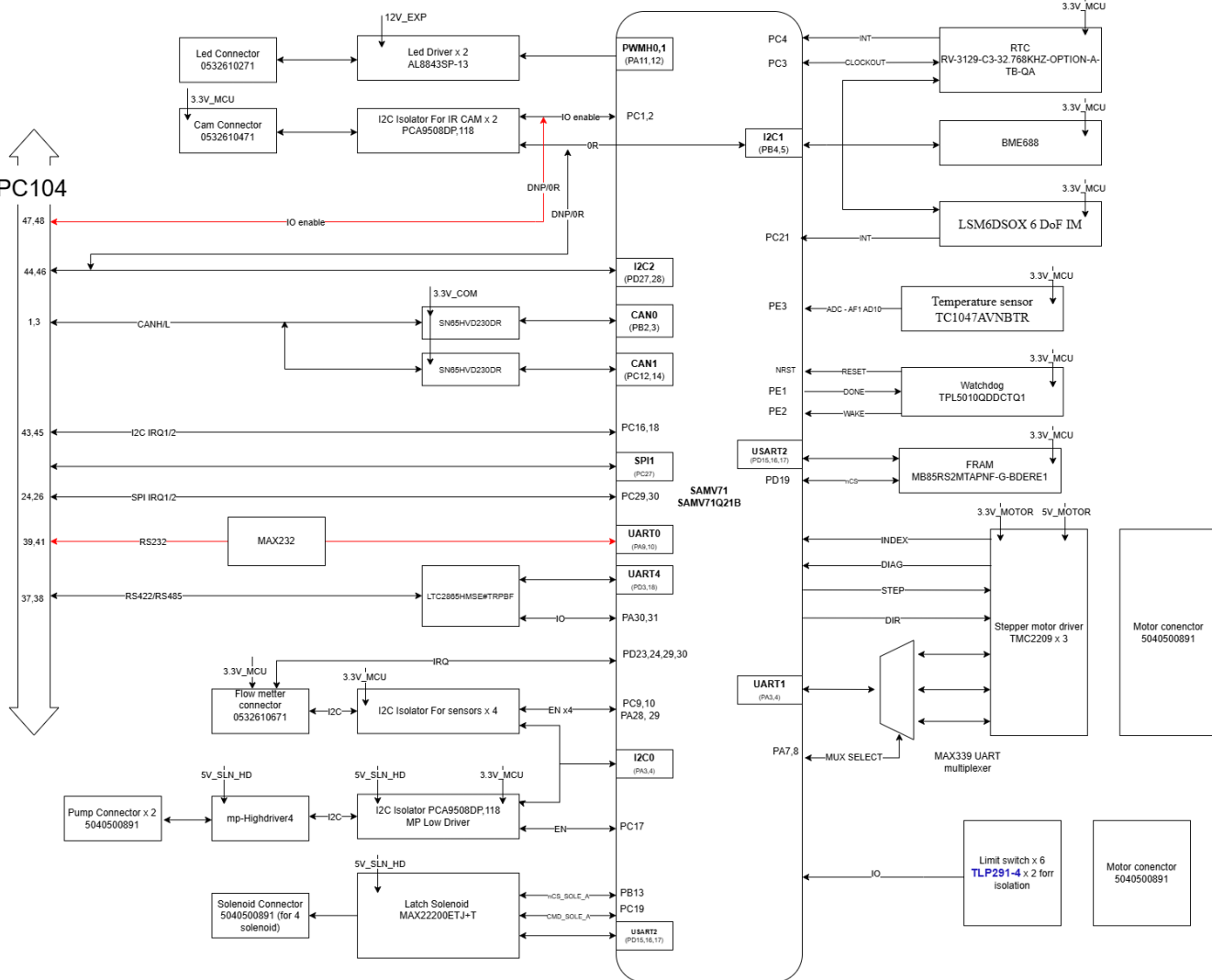
Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title: 01_INDEX.SchDoc	Sheet: 1 of 27
Checked by:	Bao Bui	Doc no: 1	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 13-Aug-26

PC104



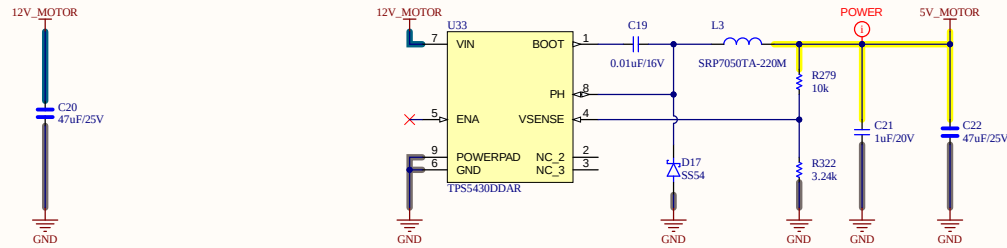
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Design by:	Duy Nguyen	Sheet title: 02_POWER_TREE.SchDoc	Sheet: 2 of 27
Checked by:	Bao Bui	Doc no: 2	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 13-Aug-26

PC104

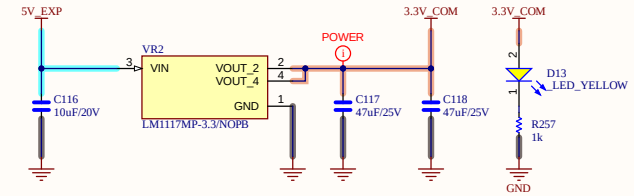


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Design by:	Duy Nguyen	Sheet title:	03_BLOCK_DIAGRAM.SchDoc
Checked by:	Bao Bui	Doc no:	3
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	3 of 27
		Size:	A3
		Date:	13-Aug-26

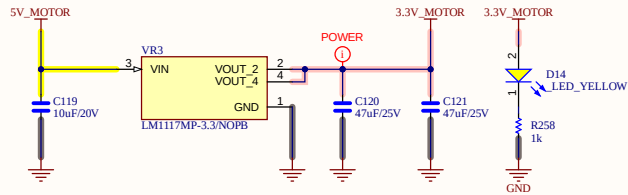
BUCK for PERIPHERAL



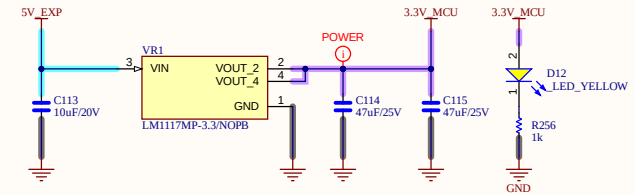
LDO for driver (driver communication)



LDO for motor



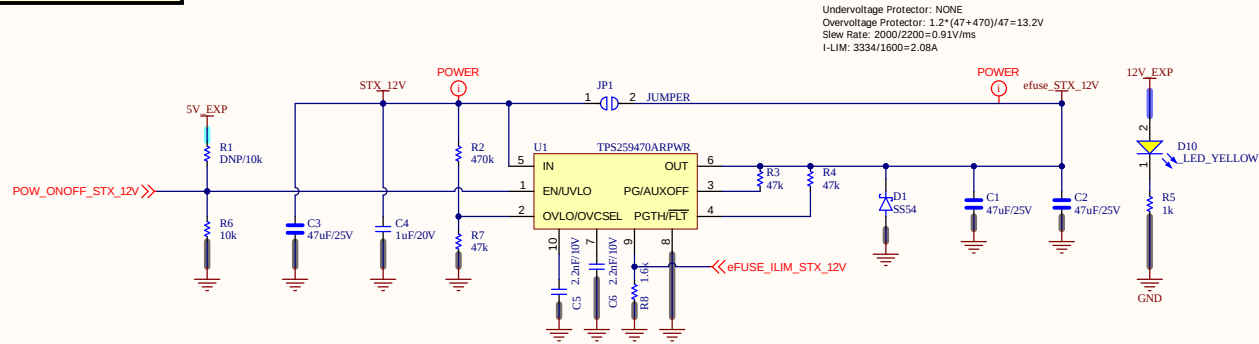
LDO for MCU



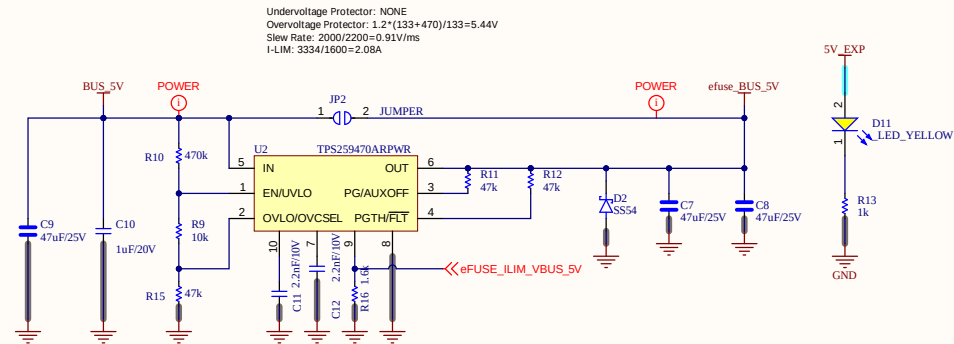
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Design by:	Duy Nguyen	Sheet title: 04_POWER.SchDoc	Sheet: 4 of 27
Checked by:	Bao Bui	Doc no: 4	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 13-Aug-26

* 2 efuse dùng cho nguồn tổng là STX 12V và BUS 5V, trong đó STX 12V sẽ có thể được MCU on/off, còn BUS 5V thì mặc định on (do MCU lấy 5V từ đây)

EFUSE STX 12V

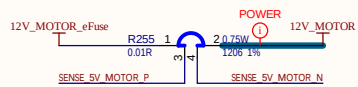
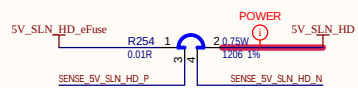
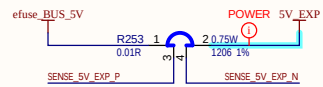
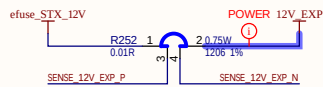
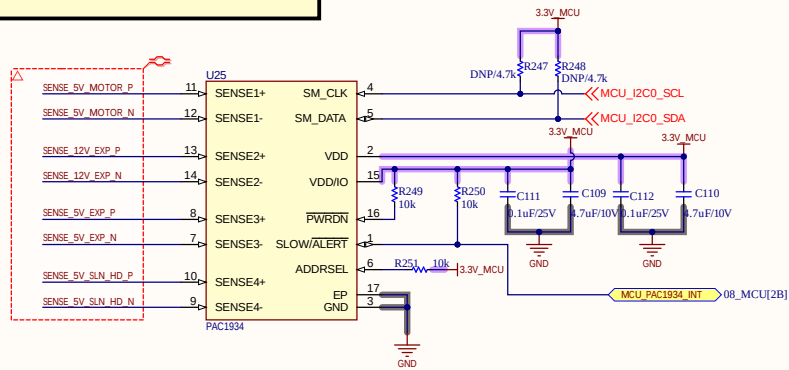


EFUSE BUS 5V



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title: 05_POWER_EFUSE.SchDoc	Sheet: 5 of 27
Checked by:	Bao Bui	Doc no: 5	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 13-Aug-26

POWER MEASUREMENT 12V AND 5V



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title: 06_POWER_MEASUREMENT.SchDoc	Sheet: 6 of 27
Checked by:	Bao Bui	Doc no: 6	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 13-Aug-26

EFUSE for SLN & HD

Undervoltage Protector: NONE
 Overvoltage Protector: $1.2 \times (133 + 470) / 133 = 5.44V$
 Sleep Rate: $2000 / 2200 = 0.91V/ms$
 I-LIM: $3334 / 1600 = 2.08A$

5V_EXP

5V_EXP

POWER

JP10

JUMPER

5V_SLN_HD_eFuse

POWER

5V_SLN_HD

D15_LED_YELLOW

TPS259470ARPWR

IN

OUT

EN/UVLO

PG/AUXOFF

OVLO/OVCSEL

PGTH/FLT

R259 DNP/10k

R264 10k

C124 47uF/25V

C125 1uF/20V

R260 470k

R265 133k

R261 47k

R262 47k

D8 SS54

C122 47uF/25V

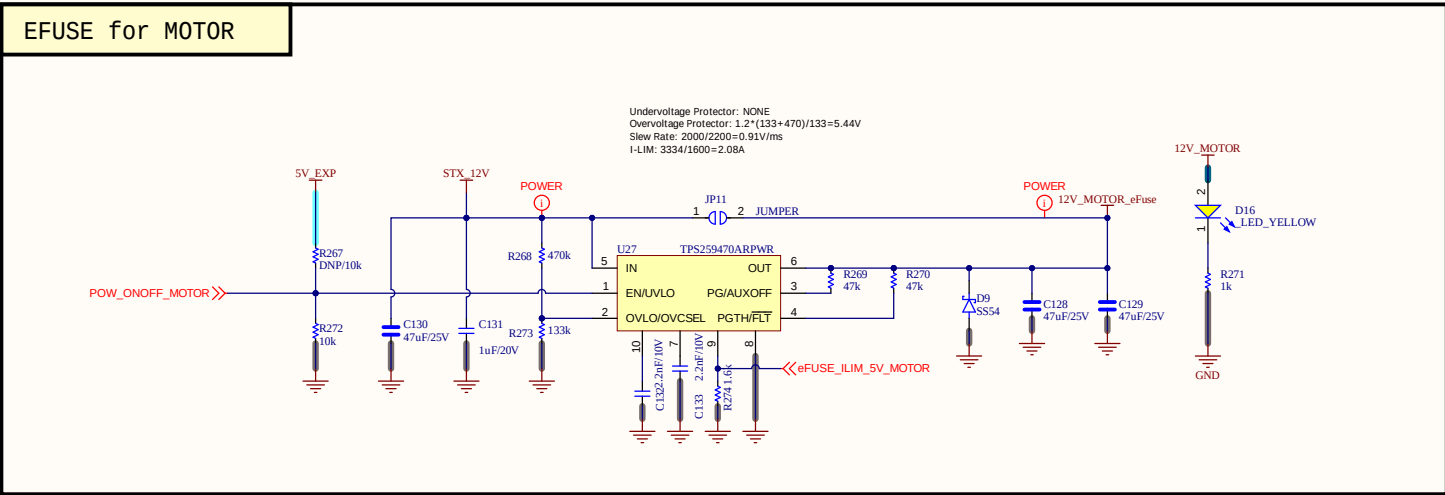
C123 47uF/25V

R263 1k

GND

POW_ONOFF_SLN_HD

«eFUSE_ILIM_5V_SLN_HD



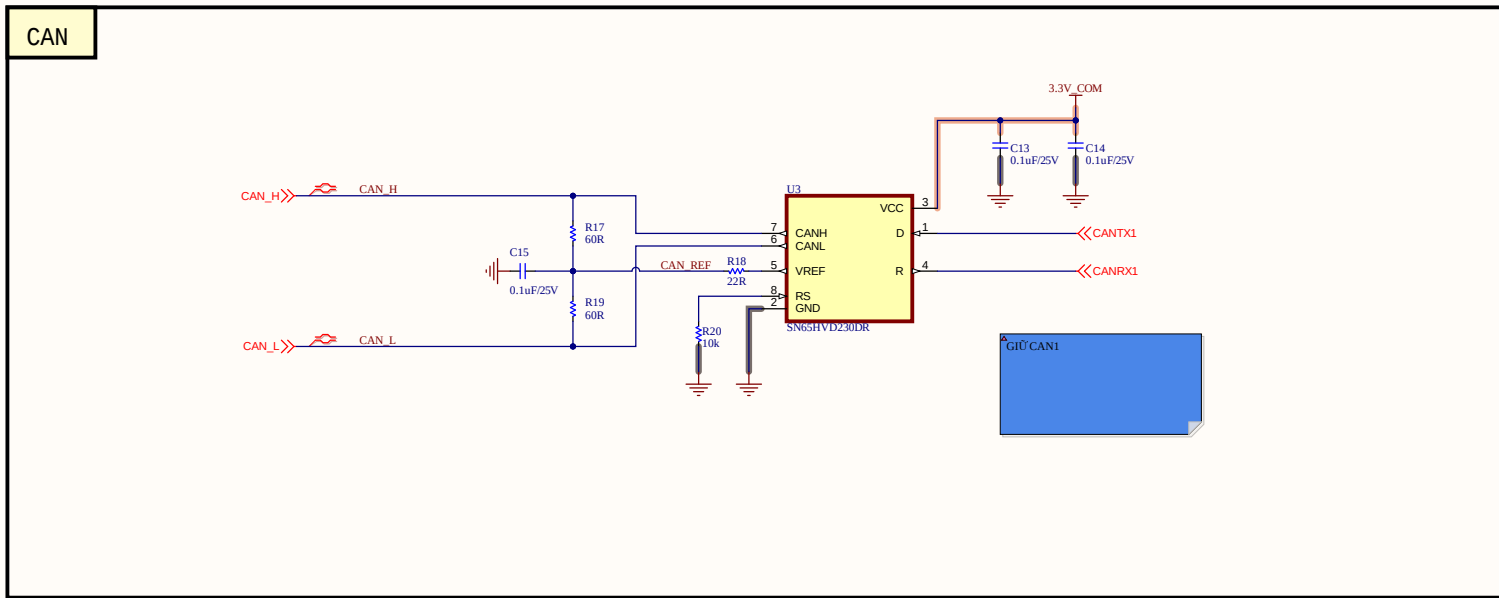
Undervoltage Protector: NONE
 Overvoltage Protector: 1.2*(133+470)/133=5.44V
 Slew Rate: 2000/2200=0.91V/ms
 I-LIM: 3334/1600=2.08A



Project	BEE_DAL	Sub-system: BEE-DAL_EXP_V1.0.PrjPcb	
Design by:	Duy Nguyen	Sheet title: 07_FUNCTION_EFUSE.SchDoc	Sheet: 7 of 27
Checked by:	Bao Bui	Doc no: 7	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 13-Aug-26

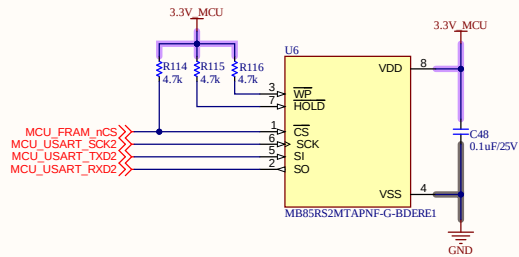


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Design by:	Duy Nguyen	Sheet title: 08_MCU.SchDoc	Sheet: 8 of 27
Checked by:	Bao Bui	Doc no: 8	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 13-Aug-26

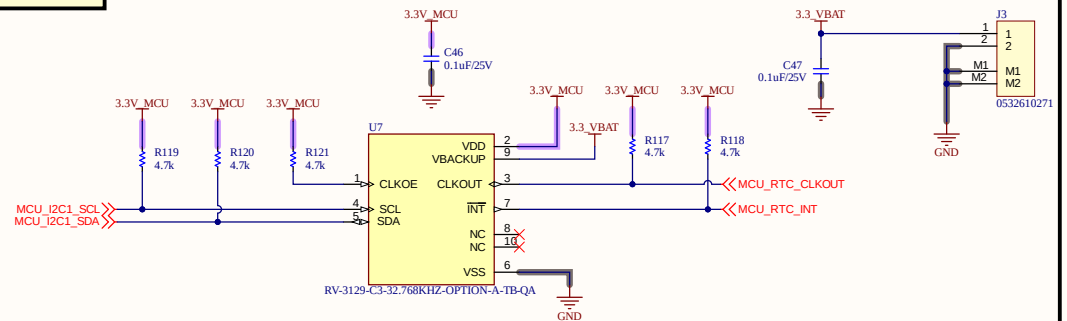


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Design by:	Duy Nguyen	Sheet title: 09_CAN.SchDoc	Sheet: 9 of 27
Checked by:	Bao Bui	Doc no: 9	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 13-Aug-26

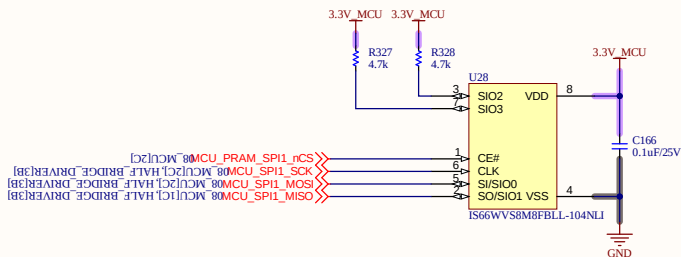
FRAM 2M



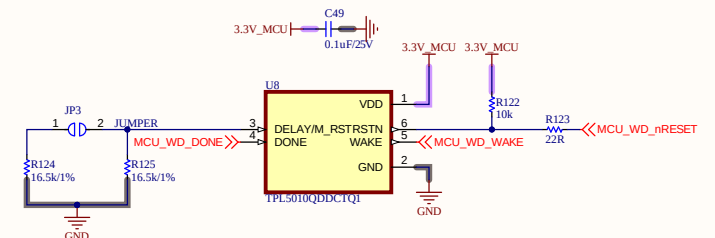
RTC



PRAM 64M

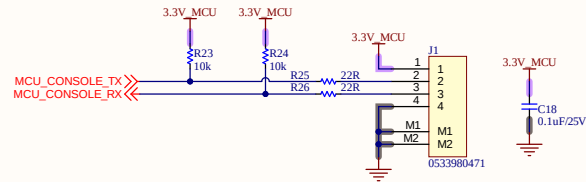


WATCH DOG

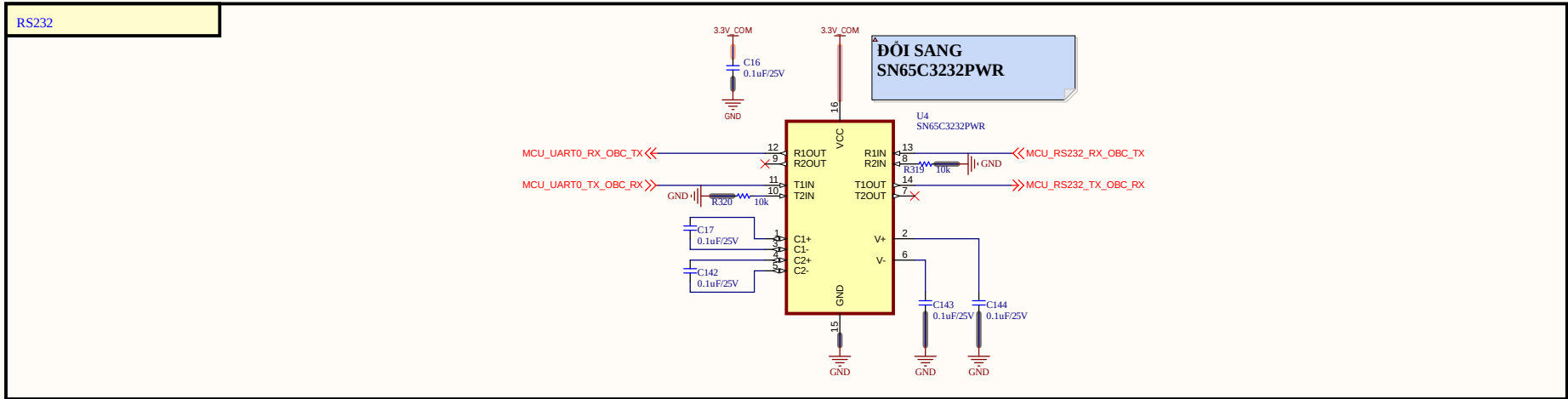
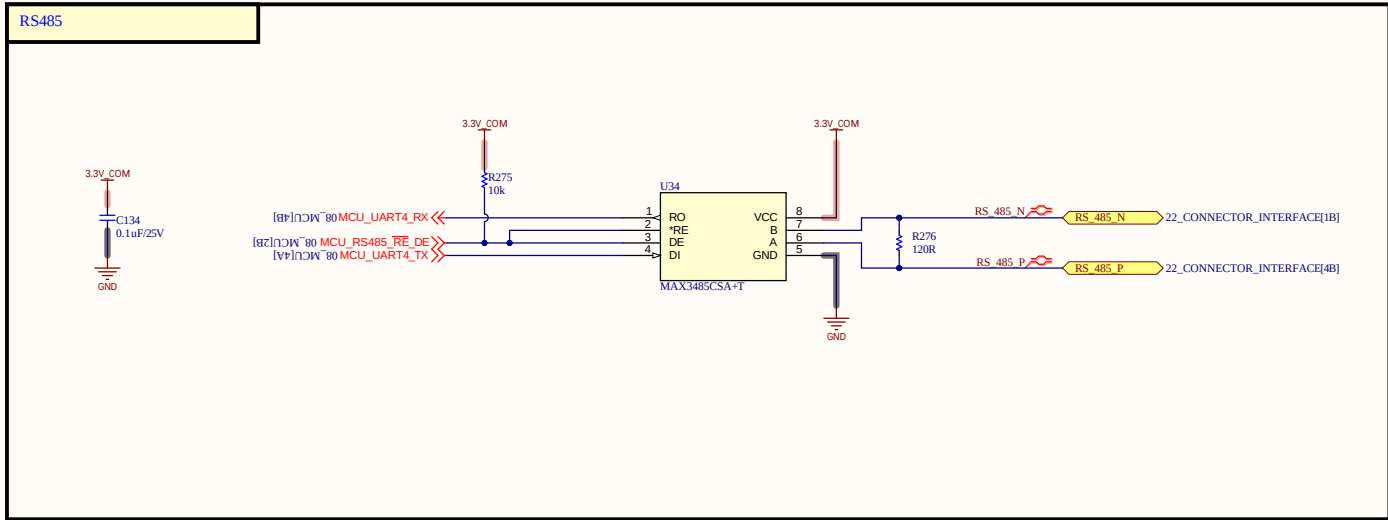


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Design by:	Duy Nguyen	Sheet title:	10_PRAM_RTC_WDT.SchDoc
Checked by:	Bao Bui	Doc no:	10
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	10 of 27
		Size:	A3
		Date:	13-Aug-26

CONSOLE

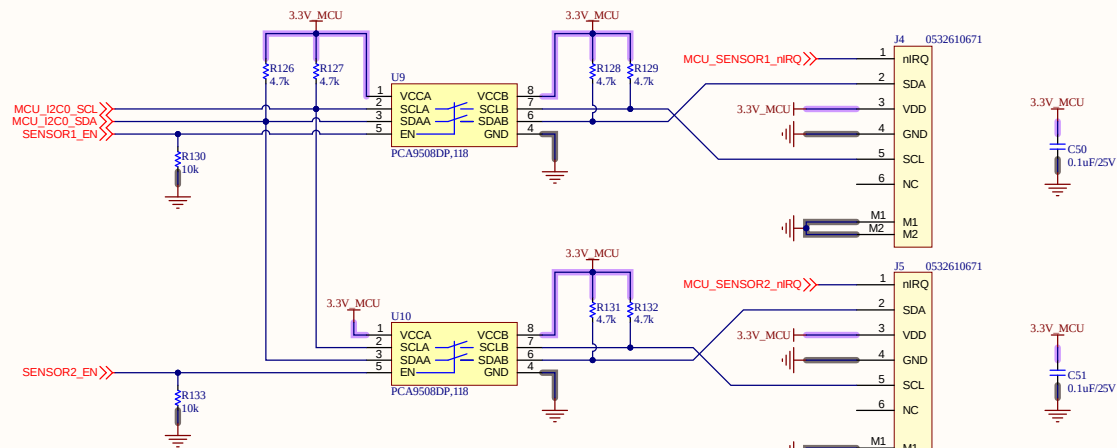


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Design by:	Duy Nguyen	Sheet title: 11_CONSOLE.SchDoc	Sheet: 11 of 27
Checked by:	Bao Bui	Doc no: 11	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 13-Aug-26



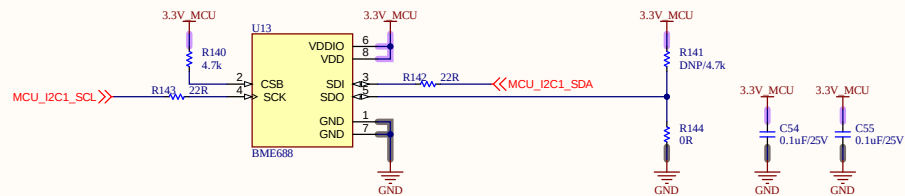
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Design by:	Duy Nguyen	Sheet title:	12_RS485_R232.SchDoc
Checked by:	Bao Bui	Doc no:	12
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	12 of 27
		Size:	A3
		Date:	13-Aug-26

I2C

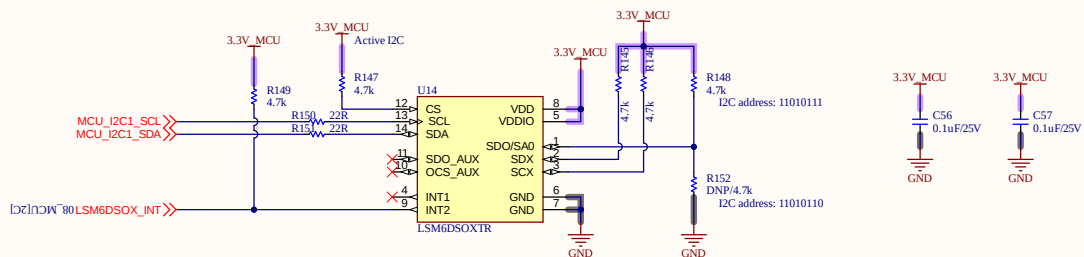


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Design by:	Duy Nguyen	Sheet title:	13_FLOW_Sensor.SchDoc
Checked by:	Bao Bui	Doc no:	13
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	13 of 27
		Size:	A3
		Date:	13-Aug-26

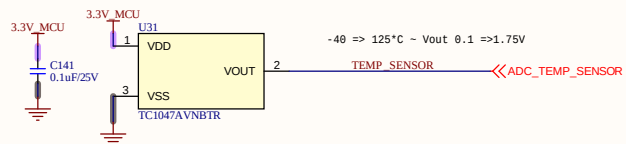
BME688



LSM6DS0X

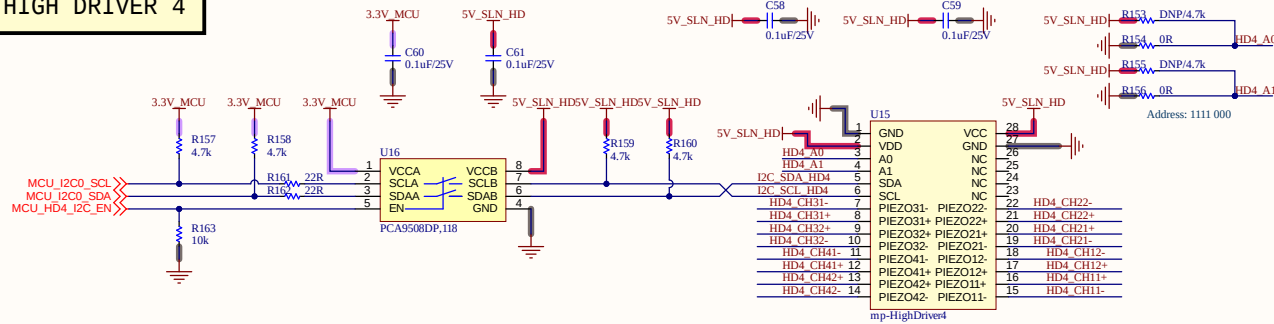


TEMPERATURE SENSOR

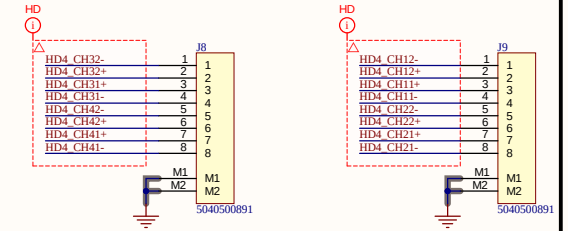


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Design by:	Duy Nguyen	Sheet title:	14_I2C_Sensor_OnBoard.SchDoc
Checked by:	Bao Bui	Doc no:	14
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	14 of 27
		Size:	A3
		Date:	13-Aug-26

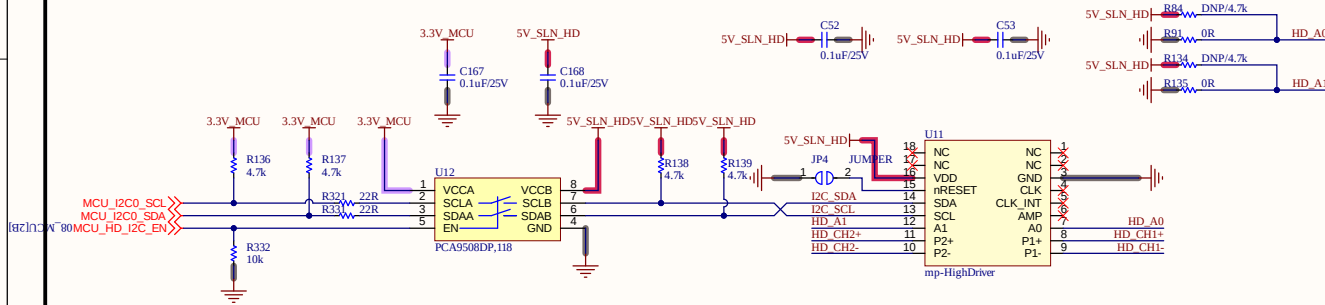
HIGH DRIVER 4



CONNECTOR BUMP

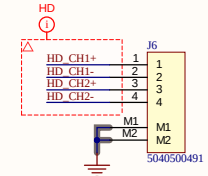


HIGH DRIVER



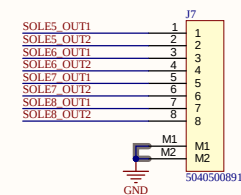
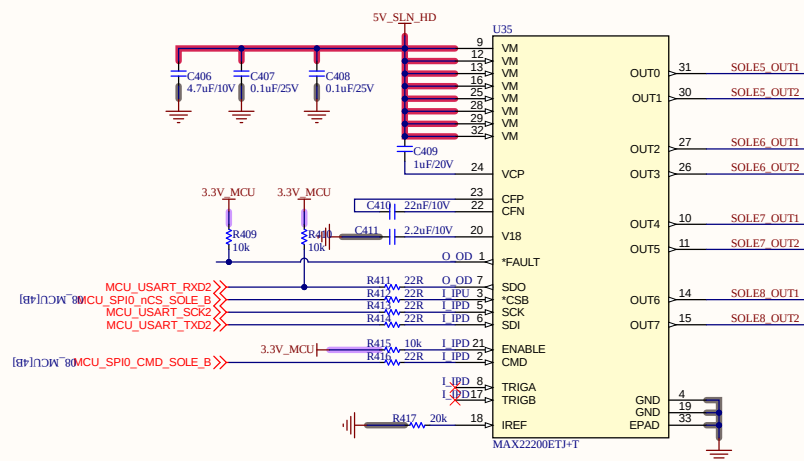
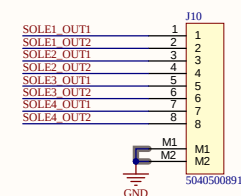
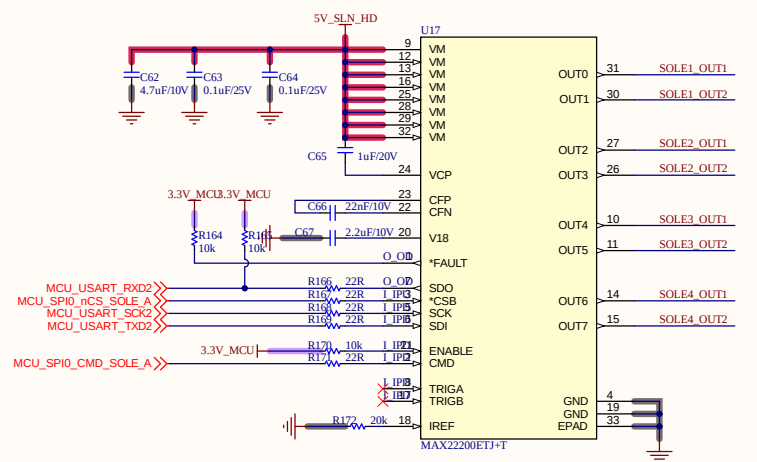
check pin lowdriver và highdriver

CONNECTOR BUMP

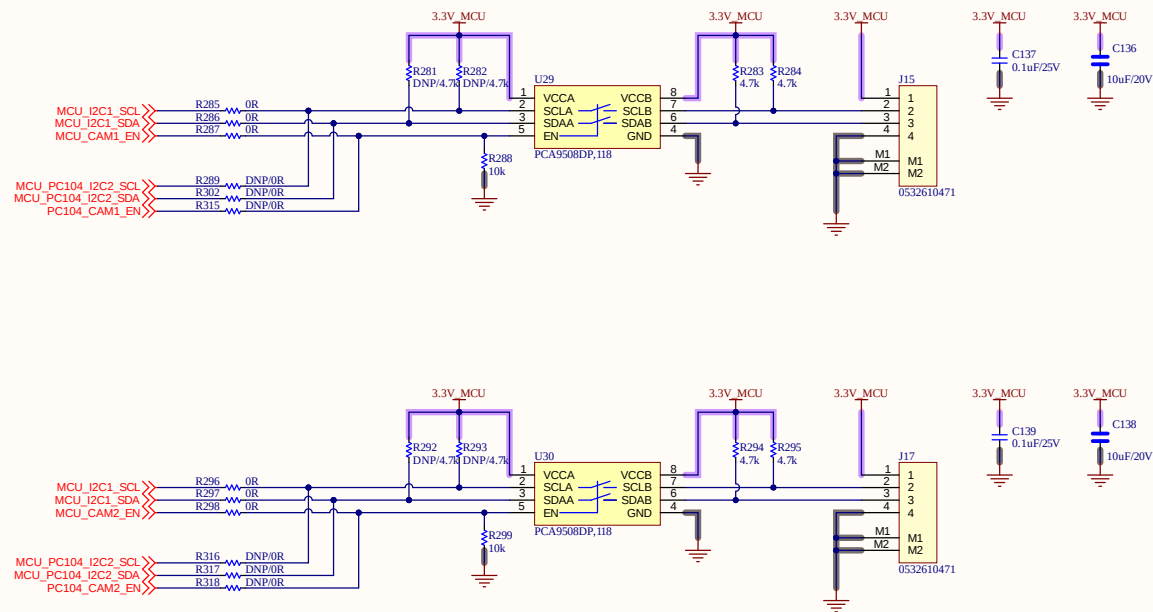


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Design by:	Duy Nguyen	Sheet title:	15_HD_PUMP.SchDoc
Checked by:	Bao Bui	Doc no:	15
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	15 of 27
		Size:	A3
		Date:	13-Aug-26

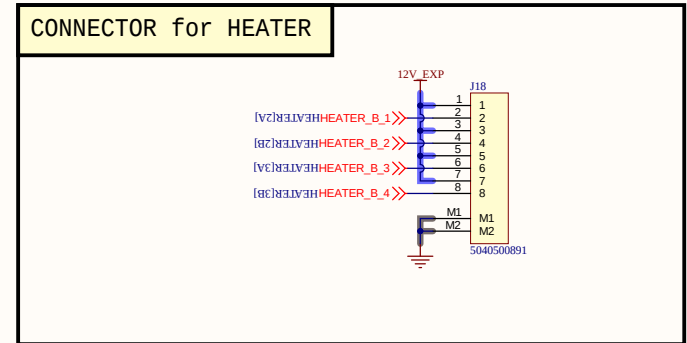
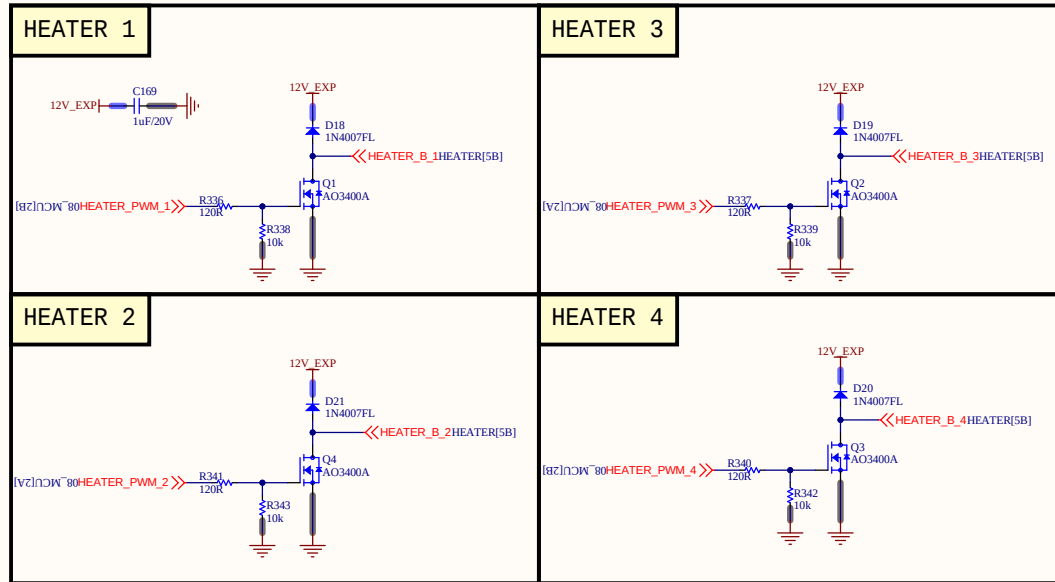
SOLENOID



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	16_SOLENOID_HB_1.SchDoc
Checked by:	Bao Bui	Doc no:	16
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	16 of 27
		Size:	A3
		Date:	13-Aug-26

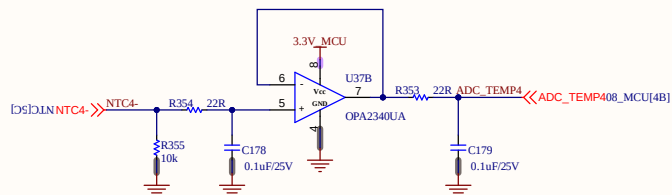
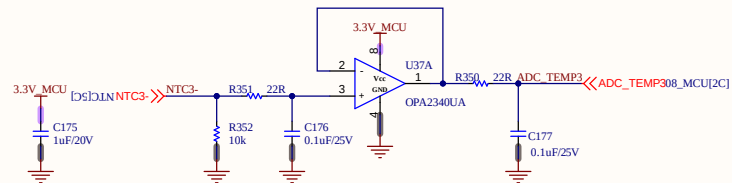
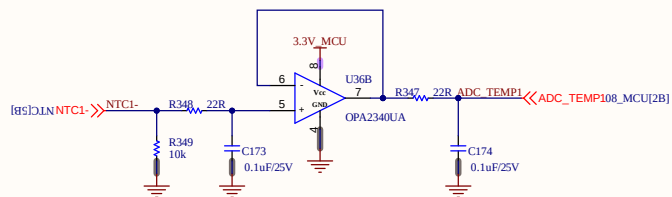
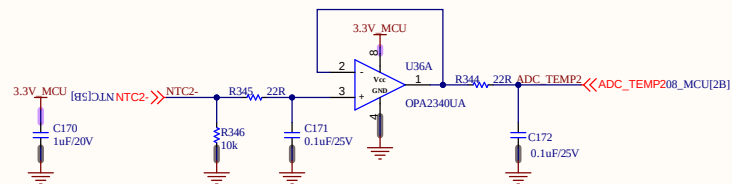


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Design by:	Duy Nguyen	Sheet title: 17_CAM_CONNECTOR.SchDoc	Sheet: 17 of 27
Checked by:	Bao Bui	Doc no: 17	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 13-Aug-26

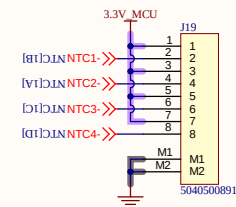


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Design by:	Duy Nguyen	Sheet title:	HEATER.SchDoc
Checked by:	Bao Bui	Doc no:	19
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	19 of 27
		Size:	A3
		Date:	13-Aug-26

NTC

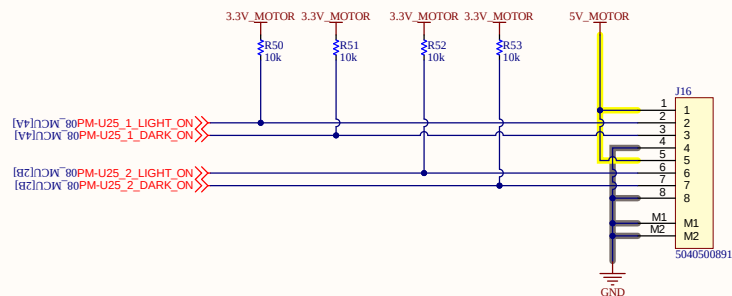


CONNECTOR FOR NTC

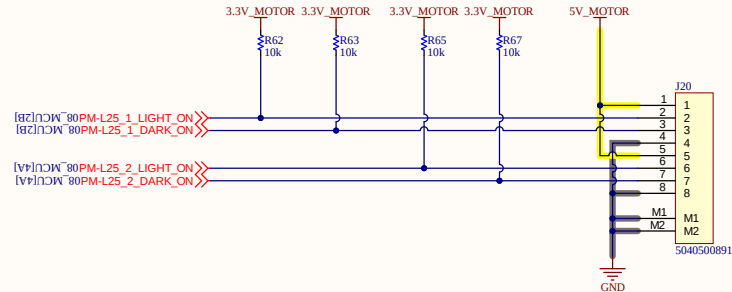


Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	NTC.SchDoc
Checked by:	Bao Bui	Doc no:	20
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	20 of 27
		Size:	A3
		Date:	13-Aug-26

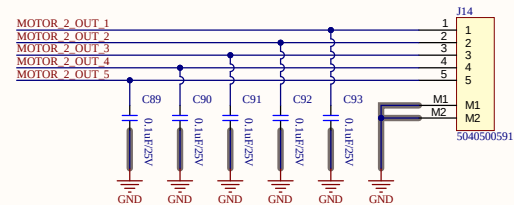
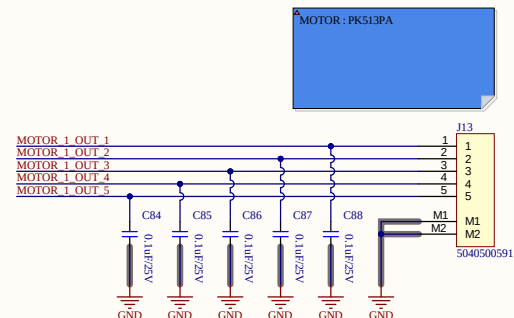
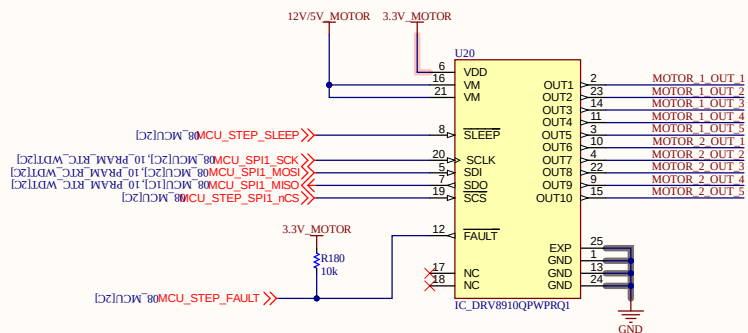
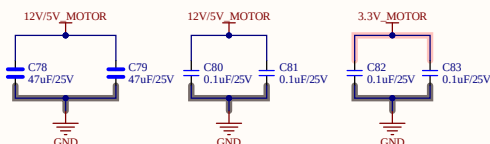
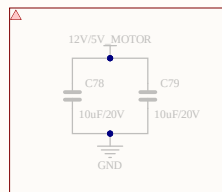
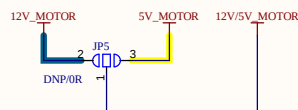
CONNECTOR for PM-U25



CONNECTOR for PM-L25

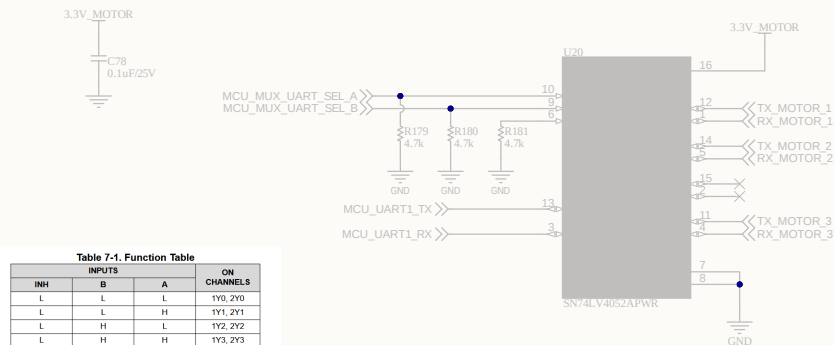


Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	LIMIT_SENSORS.SchDoc
Checked by:	Bao Bui	Doc no:	21
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	21 of 27
		Size:	A3
		Date:	13-Aug-26



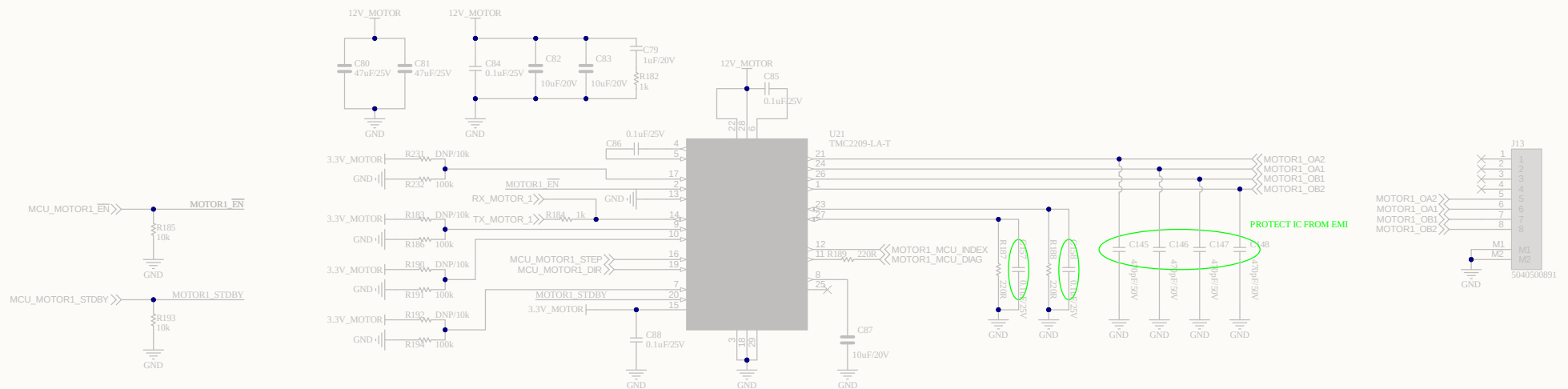
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Design by:	Duy Nguyen	Sheet title:	HALF_BRIDGE_DRIVER.SchDoc
Checked by:	Bao Bui	Doc no:	22
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	22 of 27
		Size:	A3
		Date:	13-Aug-26

UART MUX

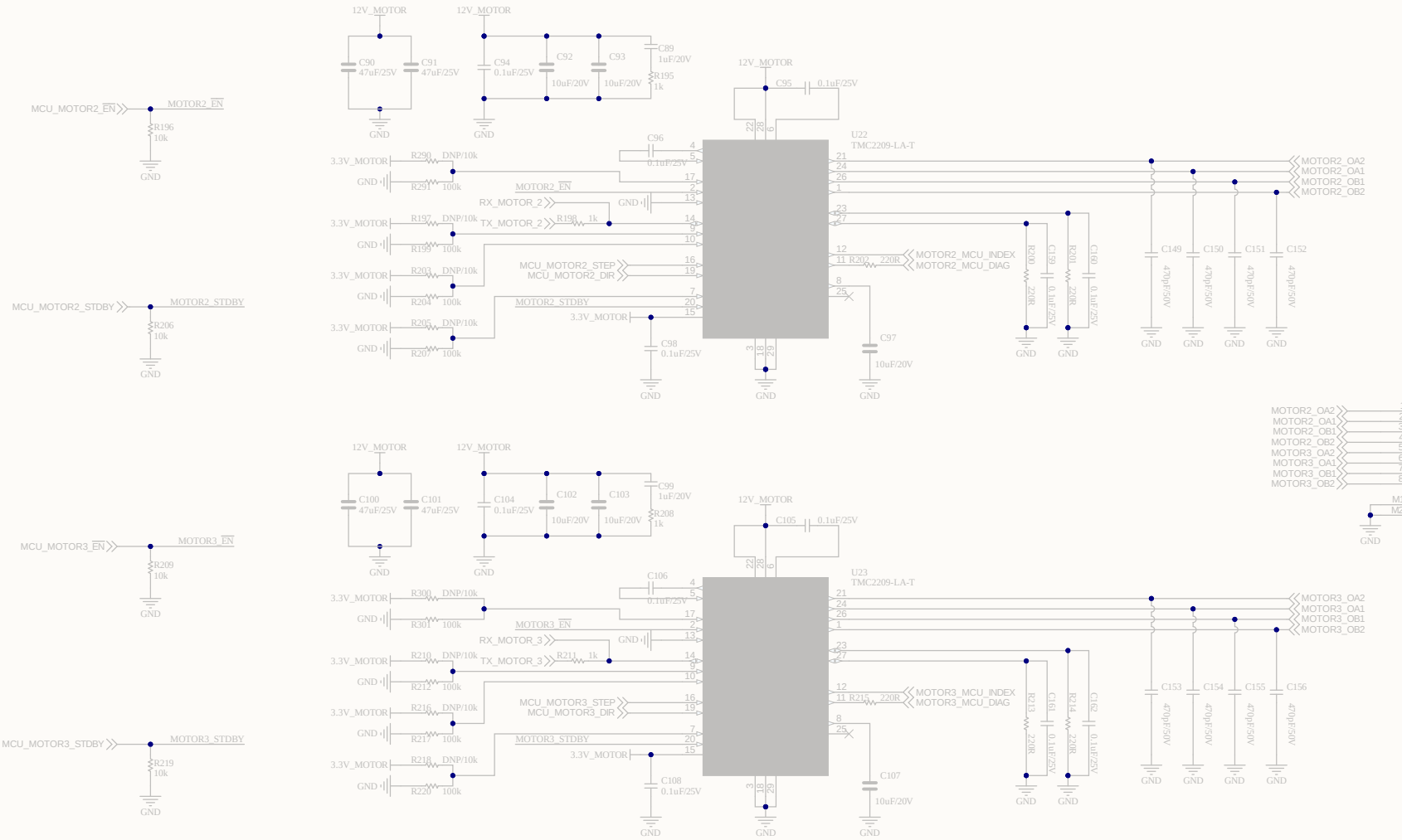


INPUTS			ON CHANNELS
INH	B	A	
L	L	L	1Y0, 2Y0
L	L	H	1Y1, 2Y1
L	H	L	1Y2, 2Y2
L	H	H	1Y3, 2Y3
H	X	X	None

INPUTS			ON CHANNELS
INH	B	A	
L	L	L	1Y0, 2Y0
L	L	H	1Y1, 2Y1
L	H	L	1Y2, 2Y2
L	H	H	1Y3, 2Y3
H	X	X	None

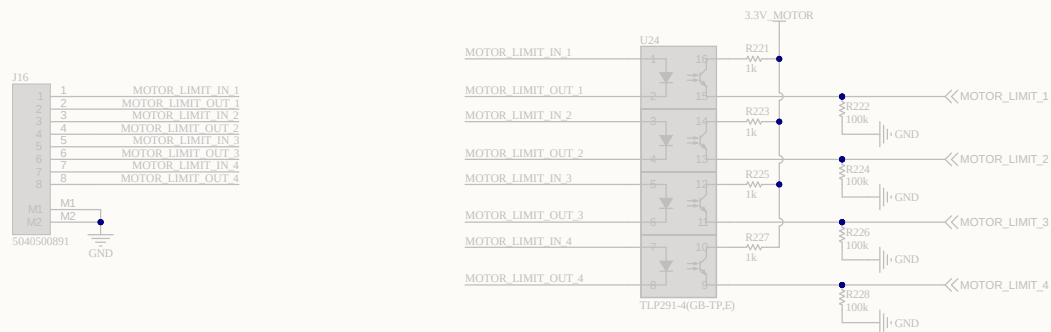


Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrgPcb
Design by:	Duy Nguyen	Sheet title:	19_STEP_MOTOR_DRIVER_1.SchDoc Sheet: 23 of 27
Checked by:	Bao Bui	Doc no:	23 Size: A3
Approved by:	Vu Pham	Revision	1.0.0 Date: 13-Aug-26

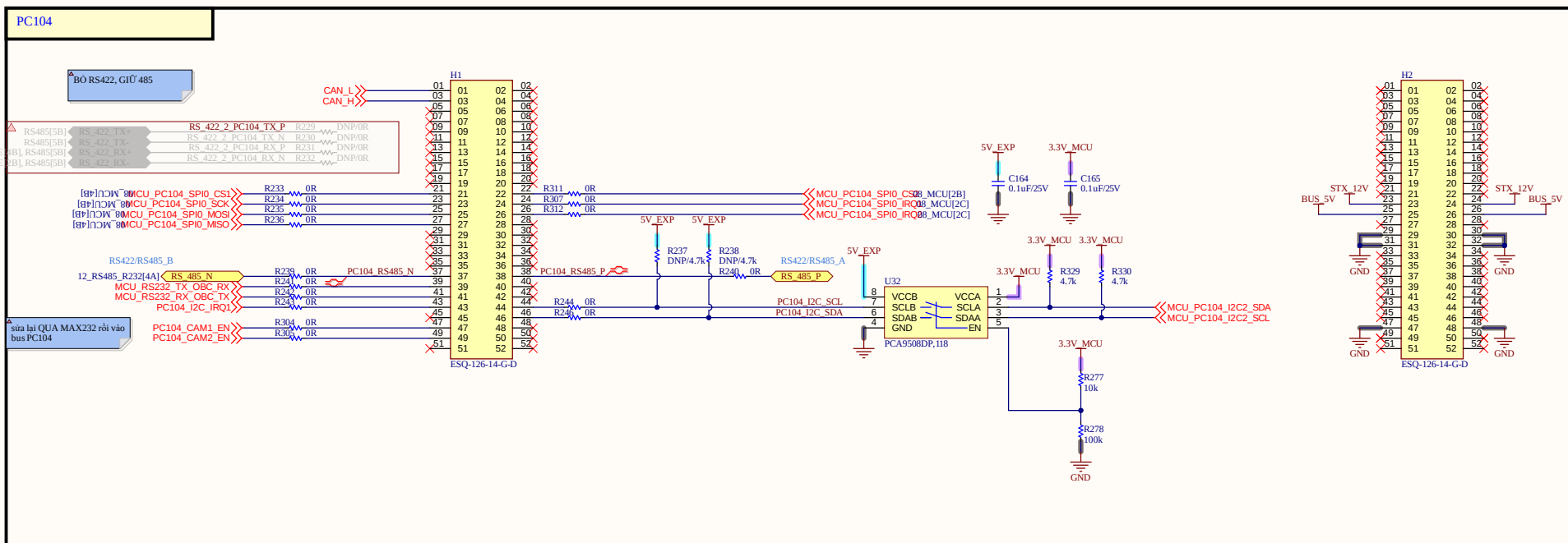


Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb		
Design by:	Duy Nguyen	Sheet title:	20_STEP_MOTOR_DRIVER_2.Sch	Sheet:	24 of 27
Checked by:	Bao Bui	Doc no:	24	Size:	A3
Approved by:	Vu Pham	Revision	1.0.0	Date:	13-Aug-26

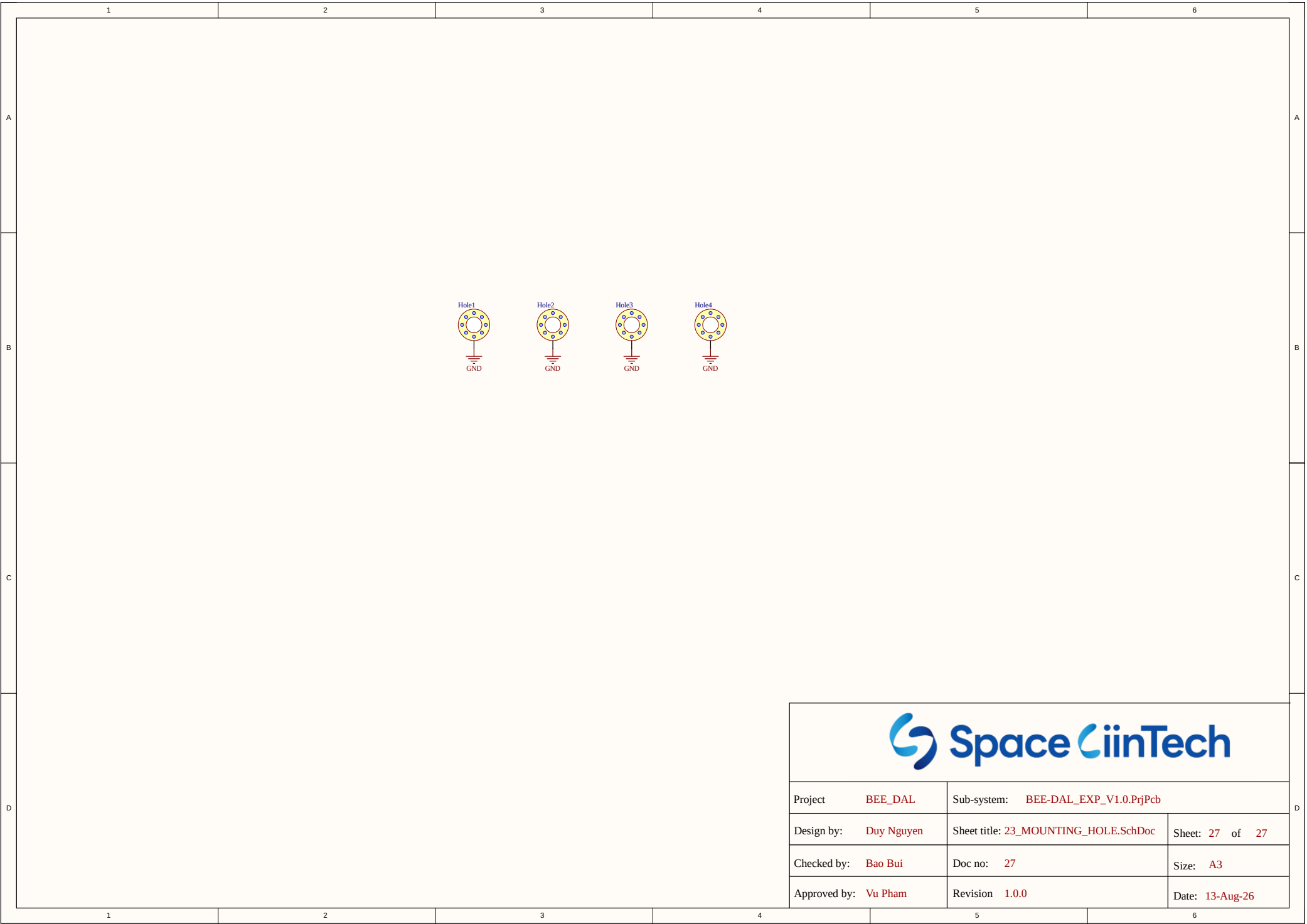
STEP LIMIT



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	21_STEP_MOTOR_ISOLATOR_INOUT_SchDoc
Checked by:	Bao Bui	Doc no:	25
Approved by:	Vu Pham	Revision	1.0.0
		Size:	A3
		Date:	13-Aug-26



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	22_CONNECTOR_INTERFACE.SchDoc
Checked by:	Bao Bui	Doc no:	26
Approved by:	Vu Pham	Revision	1.0.0
		Size:	A3
		Date:	13-Aug-26



		
Project	BEE_DAL	Sub-system: BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title: 23_MOUNTING_HOLE.SchDoc
Checked by:	Bao Bui	Doc no: 27
Approved by:	Vu Pham	Revision 1.0.0
		Sheet: 27 of 27
		Date: 13-Aug-26

